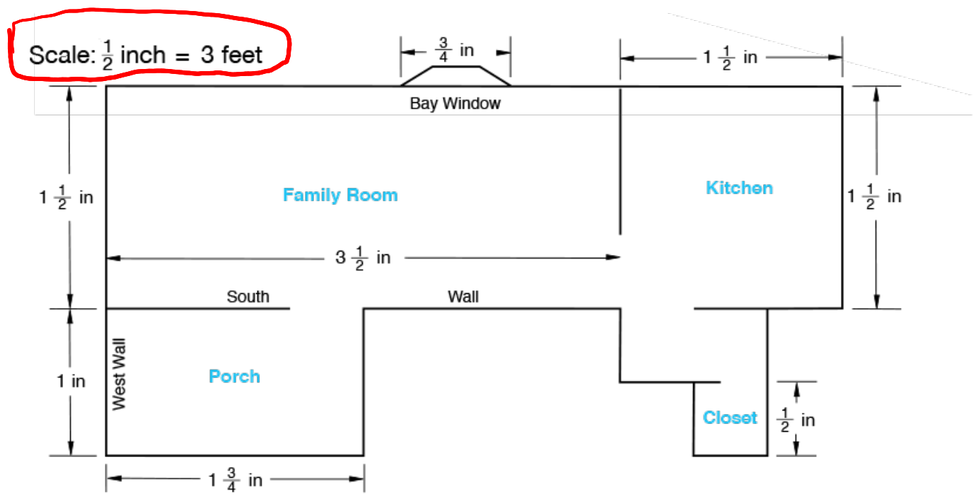
Scale Drawings

A **scale drawing** such as a floor plan or a map uses proportions to compare values in the drawing to actual values. For example, if a floor plan shows that a room's length is twice its width, then the actual room will be twice as long as it is wide.



The scale above tells you that every $\frac{1}{2}$ inch on this drawing represents 3 feet on the real floor. The scale is a ratio—inches in the drawing to feet in the building. You can determine the actual size of all the rooms by setting up a proportion.

Example

Hide

The window in the drawing measures $\frac{3}{4}$ inch. How many feet across will the bay window be?

Step 1

Set up a proportion.

$$\frac{\frac{1}{2}\text{inch}}{3\text{ feet}} = \frac{\frac{3}{4}\text{inch}}{n\text{feet}}$$

Step 2

Multiply to find the cross products and find the unknown.

$$\frac{1}{2} \times \underset{n}{\quad} = \frac{3}{4} \times 3$$

$$\frac{1}{2} \times \underset{n}{\quad} = 2\frac{1}{4}$$

$$n = \frac{1}{4}\,\frac{\mathbf{1}}{\mathbf{2}}\ \textbf{feet}$$

Step 3

Check your proportion by using cross products.

$$\frac{\frac{1}{2}}{3} = \frac{\frac{3}{4}}{4\frac{1}{2}}$$

$$\frac{1}{2} \times 4\frac{1}{2} = \frac{3}{4} \times 3$$

$$2\frac{1}{4} = 2\frac{1}{4}$$

The bay window will be $4\frac{1}{2}$ **feet across.**

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Category

Lesson 6: Ratios and Proportions

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